REE Allied Subjects: Mechanics, Thermal Science, Electronics, and Practice

REGISTERED-ELECTRICAL-ENGINEERING · engineering-sciences-and-allied-subjects · BoardPrep lesson v1

Official examination scope

PRC’s enhanced REE TOS identifies engineering sciences and allied competencies. This lesson teaches selected mechanics, thermal, electronic, and professional foundations. Consult current statutes, regulations, standards, and the relevant discipline source before applying the examples to a real installation or project.

Learning objectives

- Apply Newton’s second law to a stated system.
- Relate thermal and fluid quantities through a defined system model.
- Explain basic semiconductor and professional-practice boundaries.

1. Mechanics, free-body diagrams, and materials

For a body of known mass, Newton’s second law relates net force to acceleration, Fnet = ma. Draw a free-body diagram that includes only forces acting on the selected system, choose axes, and keep each force component in newtons. Equal and opposite interaction forces act on different bodies, so they do not cancel on one body’s diagram. Materials behavior adds stress, strain, and temperature or time effects; use properties measured for the relevant condition rather than assuming one nominal value describes every specimen.

2. Thermodynamics, heat transfer, and fluid behavior

A thermal analysis needs a system boundary, state properties, heat-transfer direction, and work terms. Energy balances track transfers and changes in state; heat is energy in transit, not a stored property. Fluid models relate pressure, density, velocity, elevation, and losses, with assumptions that vary between static fluids, idealized flows, and real networks. Check units and identify neglected terms before solving. A result from a simplified model should not be turned into an equipment rating without the appropriate data and standard.

3. Semiconductor, analog, digital, and computing foundations

Electronic circuits use device models that are valid over particular bias and frequency ranges. A semiconductor junction or amplifier is not an ideal switch or gain block under every condition. Digital logic represents states using defined voltage thresholds and timing assumptions; conversion and measurement add resolution and quantization limits. For a programming or computer-application problem, define inputs, operations, expected outputs, and error cases before choosing an implementation. Verify behavior with a transparent test rather than trusting a plausible-looking output.

4. Law, contracts, safety, and professional boundaries

Republic Act No. 7920 defines the Philippine legal framework for electrical engineering practice and professional registration; current regulations and Board issuances add detail. A contract or specification communicates scope and obligations but does not replace safety duties or the current code. Identify who has authority, what work is within competence, which document version controls, and who must approve a deviation. For safety, follow the current Philippine occupational-safety rules and project procedures; this lesson does not provide a site-specific hazard assessment or legal opinion.

Find net force from acceleration

Question: A 10 kg instrument carriage accelerates at 2 m/s². What net force acts on the carriage?
1. Identify mass m = 10 kg and acceleration a = 2 m/s².
2. Apply Fnet = ma = 10 × 2.
3. Check units: kg·m/s² = N.
Answer: 20 N in the direction of the acceleration.
Check: This is the net force; the applied force may be different when friction or other forces act.

Common misconceptions

- A force pair on interacting bodies cancels on one free-body diagram.
- A semiconductor model works outside its stated range.
- A project specification overrides a safety law or professional scope boundary.

Practice patterns

- Draw a system boundary and free-body diagram first.
- State idealizations and omitted losses.
- Check authority, version, and competence for a professional decision.

Official scope references

Official scope: Enhanced Tables of Specifications for REE — Annex A — Professional Regulatory Board of Electrical Engineering, Professional Regulation Commission — https://www.prc.gov.ph/sites/default/files/2024-40%20Annex%20A.pdf
Official scope: Program of the September 2026 Registered Electrical Engineers Licensure Examination — Professional Regulation Commission; Professional Regulatory Board of Electrical Engineering — https://www.prc.gov.ph/sites/default/files/Program%20September%202026%20Electrical%20Engineers.pdf

Instructional references

Instructional reference: University Physics Volume 1 — OpenStax, Rice University — https://openstax.org/details/books/university-physics-volume-1
Instructional reference: University Physics Volume 2 — OpenStax, Rice University — https://openstax.org/details/books/university-physics-volume-2
Instructional reference: Chemistry 2e — OpenStax, Rice University — https://openstax.org/details/books/chemistry-2e
Instructional reference: MIT OpenCourseWare 6.002 Circuits and Electronics — Massachusetts Institute of Technology OpenCourseWare — https://ocw.mit.edu/courses/6-002-circuits-and-electronics-spring-2007/
Instructional reference: Republic Act No. 7920 — New Electrical Engineering Law — Republic of the Philippines; Supreme Court E-Library — https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/3041

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